General Mathematics — Exercise Sheet 5

Function limits and continuity

Variant: Statements

Exercise 1 (A direct proof on a restricted domain). Let $f : [0, \infty) \to \mathbb{R}$ be given by $f(x) = \sqrt{x}$. Prove directly from Lecture 5's definition that f is continuous at 4 and at 0. Give an explicit δ for each $\varepsilon > 0$, without using the school continuity assumption E.

Exercise 2 (Inner values that repeatedly hit the excluded point). Define $g : \mathbb{R} \to [0, \infty)$ by $g(0) = 0$ and $g(x) = x^2 \sin^2(1/x)$ for $x \neq 0$. On $[0, \infty)$ define $p(u) = \sqrt{u}$ and define $q(0) = 1$, $q(u) = 0$ for $u > 0$. Prove $g(x) \to 0$ as $x \to 0$. Decide whether $p \circ g$ is continuous at zero and whether $q \circ g$ has a deleted limit there. Use Lecture 5's composition conditions and the school trigonometric identities.

Exercise 3 (A sequence as a domain). Let (b_n) be any real sequence. On $A = \{0\} \cup \{1/n : n \in \mathbb{N}\}$ define $f(0) = 0$ and $f(1/n) = b_n$. Prove that f is continuous at zero if and only if $b_n \to 0$. Prove that every $1/k$ is an isolated point of A and therefore a continuity point for every choice of (b_n). Use Lecture 5's relative definition and Lecture 3's convergence definition.

Exercise 4 (One continuity point among dense discontinuities). Define $F : \mathbb{R} \to \mathbb{R}$ by $F(x) = x$ for rational x and $F(x) = 0$ for irrational x. Determine exactly where F is continuous. Use Lecture 5's density argument, sequential characterisation and error bounds.

Exercise 5 (Directions, large inputs and a nested sine). Using Lecture 5, find the following limits and justify each one:

1. The left and right limits of $u(x) = x/|x|$ at zero, on $\mathbb{R} \setminus \{0\}$.
2. The limit of $v(x) = \sqrt{x^2 + 1} - x$ as $x \to +\infty$. Give an explicit large-input cutoff for every error $\varepsilon > 0$.
3. The limit of $w(x) = \sin(x \sin x)/x^2$ as $x \to 0$, with $x \neq 0$. State why the composition step is valid.